CYBERSECURITY LOG ANALYSIS REPORT

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Activity: CTM-ACTIVITY-8

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Instructions: Analyze each CSV dataset provided in the GitHub repository. Identify the likely attack type, cite specific log evidence, assess whether the attack succeeded, and explain the possible impact if the attack were successful. Do not rely on filenames or guesses; support your conclusions with the log entries.

ATTACK 1 ANALYSIS

1. Identification of Attack

SSH Brute-Force / Credential Stuffing Attack

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: BF-0001 / 2026-10-31T08:14:02Z
Observed activity: An unsuccessful SSH login attempt was detected on port 22 from source IP address 198.51.100.23, using the username admin.
Why it matters: This indicates an initial attempt to gain unauthorized access by using commonly known administrative usernames and passwords.

Evidence 2: Event ID / Timestamp: BF-0004 / 2026-10-31T08:14:17Z
Observed activity: Several unsuccessful SSH login attempts were recorded from IP address 198.51.100.23 within a 15-second period, targeting privileged accounts such as admin, administrator, and root. This activity indicates a potential brute-force attack.
Why it matters: This indicates that someone repeatedly tried different passwords in a short period to gain unauthorized access to accounts with administrative privileges.

Evidence 3: Event ID / Timestamp: BF-0005 / 2026-10-31T08:14:22Z
Observed activity: The admin account was automatically locked after five failed login attempts, helping prevent any further unauthorized access.
Why it matters: This demonstrates how the system's built-in security features automatically responded to block further login attempts and protect against unauthorized access.

3. Attack Analysis

An external attacker using IP address 198.51.100.23 attempted to access the internal server 192.0.2.10 through SSH (port 22) without authorization. The attacker repeatedly tried different passwords for common administrator accounts, such as admin, administrator, and root, making login attempts every five seconds starting at 08:14:02Z. Fortunately, the system detected the suspicious activity and took action by automatically locking the targeted account after five failed attempts (BF-0005) and activating rate-limiting by the sixth attempt (BF-0007). These security measures helped stop further login attempts and protect the server from unauthorized access.

4. Attack Status

Mitigated / Blocked (Unsuccessful)

5. Possible Scenario if the Attack Succeeds

- Set Up Automatic IP Blocking and Intrusion Prevention. Use security tools such as `fail2ban` or firewall rules to automatically block IP addresses that repeatedly attempt to log in unsuccessfully, helping protect the system from potential attacks.

6. Prevention and Mitigation

Recommendation 1: Disable SSH password authentication and require users to log in using SSH keys instead. Additionally, prevent direct root login by setting `PermitRootLogin no` in the SSH configuration. These measures will help improve server security and reduce the chances of unauthorized access.

Recommendation 2: Implement automated security measures using firewall rules to monitor repeated failed login attempts and automatically block suspicious IP addresses. This helps protect the server against brute-force attacks and minimizes the risk of unauthorized access.

Recommendation 3: Limit SSH access to trusted networks by using a firewall or VPN so that only authorized users can connect to the server. Also, require multi-factor authentication (MFA) for all remote administrative logins to provide an extra layer of protection and make it harder for attackers to gain unauthorized access.

ATTACK 2 ANALYSIS

1. Identification of Attack

Credential Phishing Attack / Social Engineering

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: PH-0002 / 2026-10-31T09:03:45Z
Observed activity: The security gateway identified a suspicious link (account-verify.example.test) in an email that appeared to come from it-support\@example.test. This raised concerns that the email might be a phishing attempt intended to trick users into revealing sensitive information.
Why it matters: The link leads to a domain that is different from the organization's official website, which may indicate that someone is pretending to represent the organization to trick recipients into clicking the link and potentially stealing their personal or sensitive information.

Evidence 2: Event ID / Timestamp: PH-0003 / 2026-10-31T09:05:18Z
Observed activity: The employee (employee01\@example.test) clicked the suspicious link in the phishing email, which may have exposed their account or sensitive information to potential security risks.
Why it matters: This indicates that the phishing email successfully fooled the employee into clicking the malicious link by using a fake password expiration warning to create a sense of urgency and pressure them into taking action.

Evidence 3: Event ID / Timestamp: PH-0004 / 2026-10-31T09:05:41Z
Observed activity: The browser or endpoint security system identified a fake login page on account-verify.example.test and immediately blocked it, helping prevent the employee from entering their username, password, or other sensitive login information.
Why it matters: This confirms that the link led to a fake login page created to trick employees into entering their company usernames and passwords, potentially allowing attackers to steal their login credentials.

3. Attack Analysis

An attacker using the IP address **203.0.113.80** sent a phishing email to **employee01@example.test**, posing as a member of the organization's IT support team. The email, with the subject **"Urgent: Password Expiration,"** attempted to scare the employee into providing their login credentials by claiming that their account would soon be disabled. At **09:05:18Z**, the employee clicked the suspicious link and was redirected to **account-verify.example.test**. Fortunately, the browser's security protection recognized the page as a fake login site and blocked it before the employee could enter any credentials (PH-0004). In addition, the email gateway successfully quarantined the phishing email (PH-0005), helping prevent further exposure to the threat.

4. Attack Status

Mitigated / Blocked (Unsuccessful)

5. Possible Scenario if the Attack Succeeds

The attacker manages to obtain the employee's corporate username and password through the fake login page, which could potentially be used to gain unauthorized access to the organization's systems and accounts.

6. Prevention and Mitigation

Recommendation 1: Provides an additional layer of protection by ensuring that even if attackers manage to steal a password, they still cannot easily access the company's systems without completing another authentication step.

Recommendation 2: Helps detect and block suspicious emails that pretend to come from internal company addresses, reducing the chances of employees falling victim to phishing attacks.

Recommendation 3: Conduct regular security awareness training and simulated phishing activities to help employees recognize suspicious emails, spot fake or unfamiliar websites, avoid clicking urgent or misleading links, and promptly report possible phishing attempts to the IT or security team.

ATTACK 3 ANALYSIS

1. Identification of Attack

Distributed Denial of Service (DDoS) / HTTP Flood Attack

2. Evidence from the System Logs

Evidence 1: Event ID / Timestamp: DOS-0002 / 2026-10-31T10:20:10Z
Observed activity: Traffic from IP address 198.51.100.90 suddenly spiked to 420 requests per second, which was significantly higher than the normal rate of 35 requests per second. The system recognized this unusual activity and automatically applied HTTP 429 rate limiting to control the excessive requests and help prevent possible service disruptions.
Why it matters: This suggests that the attacker is sending an unusually large number of requests to overwhelm the server and use up its resources, which could slow down or interrupt services for legitimate users.

Evidence 2: Event ID / Timestamp: DOS-0004 / 2026-10-31T10:20:30Z
Observed activity: Traffic from several IP addresses suddenly surged to 850 requests per second, far exceeding the normal rate of 35 requests per second. As a result, the service became slower and started returning HTTP 503 (Service Unavailable) errors, showing that the server was having difficulty handling the unusually high volume of incoming requests.
Why it matters: This confirms that traffic from multiple sources overwhelmed the server's capacity, causing its performance to slow down and preventing legitimate users from accessing the service properly.

Evidence 3: Event ID / Timestamp: DOS-0005 / 2026-10-31T10:20:40Z
Observed activity: The automated health check was unable to pass during the ongoing traffic spike because the service kept returning HTTP 503 errors. This indicated that the service was temporarily down or struggling to process the large number of incoming requests.
Why it matters: This confirms that the web application became unavailable because the massive amount of incoming traffic exhausted its resources, making it difficult for the server to process and respond to legitimate user requests.

3. Attack Analysis

At **10:20:00Z**, an attacker launched a coordinated HTTP flood attack against the web server at **192.0.2.20** through HTTPS port **443**. Traffic suddenly increased from the normal rate of 35 requests per second to 850 requests per second, coming from multiple IP addresses (**198.51.100.90–198.51.100.93**). The overwhelming amount of traffic caused the server to slow down significantly and return HTTP 503 errors. As a result, the automated health check failed at **10:20:40Z** because the server could no longer respond properly to incoming requests. To address the attack, the security system automatically enabled traffic filtering and rate limiting at **10:20:50Z** (DOS-0006), reducing the incoming traffic to 160 requests per second. These security measures helped stabilize the server and successfully restore normal service operations by **10:22:00Z** (DOS-0008).

4. Attack Status

Mitigated / Resolved

5. Possible Scenario if the Attack Succeeds

The web application may become inaccessible or stop responding, making it difficult for legitimate users to use the service and potentially causing the organization to fall short of its promised service availability and uptime targets.

6. Prevention and Mitigation

Recommendation 1: Implement a Web Application Firewall (WAF) and cloud-based DDoS protection services, such as Cloudflare or AWS Shield, to identify and block harmful traffic before it reaches the main servers. These security measures help prevent large-scale traffic attacks, keep the application running smoothly, and ensure that legitimate users can continue accessing the system without interruption.

Recommendation 2: Set up edge proxies to automatically control how many requests each IP address or network can send within a certain period. When unusual activity is detected, suspicious traffic from untrusted IP ranges can also be blocked or restricted. This helps prevent excessive traffic from overwhelming the server, minimize malicious requests, and ensure that legitimate users can continue accessing the application without interruption.

Recommendation 3: Implement load balancers and automatic scaling to distribute incoming traffic across multiple servers and automatically increase server resources when there is a sudden rise in legitimate user activity. This helps maintain system performance, prevent server overload, and keep the application running smoothly.

FINAL REFLECTION

1. Among the three attacks, which could cause the greatest damage to an organization? Explain your answer.

In my opinion, Attack 2 could cause the most serious damage to an organization. Even with strong cybersecurity measures in place, there is still a chance that employees could fall for phishing attacks, especially those who are not familiar with common cybersecurity threats. This is why I recommend conducting regular security awareness training and simulated phishing exercises. These activities can help employees become more aware of suspicious emails, recognize fake or unfamiliar domains, avoid clicking on urgent or misleading links, and report possible phishing attempts to the IT or security team.

2. Why is analyzing system logs important in detecting and responding to cybersecurity incidents?

Analyzing system logs is essential because it helps organizations monitor and understand the activities happening within their systems. These logs contain important information, such as login attempts, access requests, system errors, and unusual network activity. By regularly reviewing them, security teams can detect suspicious behavior, recognize possible cyberattacks, and determine how an attacker gained access and what impact the attack had on the system. This helps organizations respond to security threats more quickly, prevent further damage, and strengthen their security measures to avoid similar incidents in the future.

3. What is the most important cybersecurity lesson you learned from this activity?

These three attacks show how important defense-in-depth and automated security measures are in protecting an organization when unexpected threats occur. Each attack targeted a different weakness: brute-force attempts to gain access through SSH, phishing emails that trick employees into revealing their login credentials, and massive amounts of malicious traffic intended to overwhelm a web application. What I find most important is that even when an employee made a mistake by clicking a suspicious phishing link, it did not lead to a major security breach. This was because other security measures, such as browser protection, automatic account lockouts, and traffic filtering, were able to detect and stop the threats before they caused further damage. Overall, effective cybersecurity is not about expecting employees or systems to be perfect. It is about having multiple layers of protection in place so that when one security measure fails or someone makes a mistake, the other defenses can help prevent the entire organization from being compromised.

SUBMISSION CHECKLIST

☐ I analyzed all three CSV datasets.

☐ I identified each attack using evidence from the logs.

☐ I included at least three supporting log observations for each attack.

☐ I justified the status of each attack.

☐ I explained a possible successful-attack scenario for each attack.

☐ I provided at least three relevant security recommendations for each attack.

☐ I completed the final reflection.

Submission filename: Surname_Firstname_Cybersecurity_Log_Analysis.docx

Reminder: Distinguish between an attack attempt, detection, prevention, and actual compromise. A suspicious event or a clicked link does not automatically prove that an account or system was compromised. Base your conclusion on the available evidence.